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One hundred years of the design of experiments on and off the pages of *Biometrika*

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Summary

The earliest important papers on planned experiments in *Biometrika* appeared within a year of each other in 1917 and 1918. Roughly speaking, one was concerned with design optimality and industrial experiments, the other with agricultural trials and blocking. We find this approximate division of the subject into two parts helpful in describing its development and growth. As a result of the hostility between Fisher and Karl Pearson, much of the development of designs for agriculture in the 1920s and 1930s occurred off the pages of *Biometrika*. Despite this, we are able to trace a coherent history of the development of the subject from field trials and response surface methods to clinical trials and Bayesian versions of design optimality.

Some key words: Clinical trial; Factorial experiment; Incomplete block design; Neighbours; Optimum design; Randomisation; Response surface methods.

1. What is design?

The majority of the papers in *Biometrika* on the design of experiments fall into two categories, roughly those for agricultural experiments and those for industrial experiments; there are also some papers on the design, rather than the analysis, of clinical trials. In the most extreme type of 'agricultural' experiment, units are physically distinct, with a probably complicated structure of errors associated with units, plots and blocks. In such experiments the treatments are typically discrete, with little or no structure. Industrial experiments are often also of this type, for example Tippett's use of a Graeco-Latin square to investigate defective cotton spindles, described by Cox (1958b, p. 213). However, many 'industrial' experiments are the reverse of this; the units can be just another portion of chemical reagents, with no specific properties. The error structure is taken as being simple, but the treatment structure may be complicated, the setting of several continuous variables. Accordingly, we use the term 'industrial' to describe the distinct methodological features in design introduced by such experiments. Much of the interesting theoretical work in experimental design is concerned with working in from either side of this gap, but the basic divide is that between experiments for which, for much of our period, the analysis

of variance would have been described as the appropriate method of analysis and those for which regression would have been considered.

The procedure of designing an experiment consists of three important phases: (a) choice of treatments; (b) choice of experimental units; and (c) deciding which treatment to apply to which experimental unit. The relative importance of the three phases depends on the application.

In agricultural and medical work it is frequently impossible to obtain a sufficient number of experimental units which are all alike, so some sort of blocking must be considered. This gives structure to the experimental units. In a clinical trial, should patients be blocked by sex or by age or by other relevant factors? In this application, blocks are often called 'strata'. Or should a cross-over design be used? Then the experimental unit becomes a patient-period. In a calf-feeding trial, is it better to use eight pens of 15 calves each or twelve pens of 10? The answer depends partly on whether treatments can be applied to, and responses measured on, individual calves or whole pens.

As soon as there is structure, phase (c) becomes important. Different ways of allocating treatments may lead to differences in the variances of estimators. In this context the 'design' is the function allocating treatments to plots.

When all treatment factors are qualitative, choice of treatments is important but not mathematically sophisticated. A clinical trial may compare one new drug with a placebo; a variety trial may compare all those varieties which passed some hurdle in last year's trials. In agricultural research, even quantitative factors may not have their potential levels throughout some interval in $\mathbb{R}$; for example, a farmer is unlikely to split bags of fertiliser into complicated fractions.

Choice of treatments becomes much more complicated when factors are genuinely quantitative, that is, having potential levels throughout an interval I of $\mathbb{R}$. Since one cannot apply all potential levels, the experimenter is immediately faced with a modelling problem. Supposing that the expected response comes from a suitably finite subset M of $\mathbb{R}^I$, the experimenter must choose a finite subset $\mathscr{X}$ of I to use as treatments in such a way that the expected response in M can be well fitted and estimated.

On people and animals, the most common type of experiment with genuinely quantitative factors is the dose–response study to estimate a critical quantile: what quantity of this poison will kill 95% of rats? Such studies, as reported in the literature, do not seem to be concerned with any structure on the experimental units. Experiments on industrial processes often have genuinely quantitative factors: it is usually claimed that there is no problem about homogeneity of experimental units. Statisticians designing such experiments concentrate on phase (a) and ignore (b) and (c). For them, the 'design' is the chosen subset $\mathscr{X}$ of I, or of $I_1 \times I_2 \times \ldots \times I_n$ if there are n treatment factors.

In many clinical trials, patients enter the trial sequentially. Thus phase (b) becomes important, and phase (c) is complicated by lack of foreknowledge of the block structure. These do not really fit into our 'agricultural'–'industrial' dichotomy.

2. DESIGN IN *BIOMETRIKA*

Examples of both kinds of design occur early in *Biometrika*.

The first *Biometrika* paper to consider design was by W. S. Gosset, writing as "Student" (1917). He commented that, when the experiment is the laboratory determination of the chemical content of biological substances, there are differences between days. He recommended assuming that the variance components structure is that of tests nested within

days, and almost suggested that groups of consecutive days should be used as blocks. He expanded these remarks in "Student" (1927). In "Student" (1923) he turned his attention to variety trials, pointing out the need to design and analyse such trials to allow for spatial trend and for positive neighbour correlation, but see § 4. In "Student" (1931) he criticised the infamous Lanarkshire milk experiment both for its lack of objective randomisation and for confounding an important treatment difference with the difference between schools. He was in correspondence with Fisher throughout this period, and quoted parts of Fisher's letters in his papers.

Gosset's emphasis on the effect of the ordering of units in time is astonishing given the continuing scant attention in the literature on the design of industrial experiments to the effect of time or of the order in which factors are changed. If, in a factorial experiment, complete randomisation is rejected because some factor levels are hard to change, split-plotting may inadvertently be introduced. Some examples are given in the introduction to Draper & John (1998); see also § 10·1.

Smith (1918), in an amazing paper which was 30 years before its time, calculated what, in § 6, are called G-optimum designs. These designs are examples of those found by the optimum design theory of Kiefer. The model is assumed known, in Smith's paper polynomials of order up to 6 in a single variable x. The design region $\mathscr{X}$ is also known, with x between -1 and 1. Finally, the observational errors are independent, identically distributed and additive, so that ordinary least squares is the appropriate estimation method. For this well-specified situation she obtained the designs which minimise the maximum variance of prediction for each polynomial over the design region. The optimum designs were not only found, but also proved to be optimum, and other error distributions and designs were considered, for example uniform designs. As a result, her paper was even longer than this one.

Despite her surname, Kirstine Smith (1878–1939) was a Danish pupil of Thiele who worked at Karl Pearson's laboratory (Hald, 1998, p. 712). She inadvertently had a greater effect on the history of statistics and the development of our subject than would be suggested by her design paper. Her earlier *Biometrika* paper, Smith (1916), discussing minimum chi-squared estimation, was criticised by Fisher. According to Egon Pearson (1968, p. 456) Karl Pearson twice rejected Fisher's criticism, giving as an explanation 'I must keep the little space I have in *Biometrika* free from controversy'. We mention the consequences for *Biometrika* in § 3.

Kirstine Smith did not extend her work on experimental design. Her third, and last, *Biometrika* paper (Smith, 1922) was on genetics. In 1924 she retired from scientific work and became a schoolteacher 'because she felt a need to work more closely with people'. Fuller biographical details are available at www.webdoe.cc. We are grateful to Professor S. Lauritzen and Dr S. B. Crary for this information.

Smith's design paper seems not to have had much impact. Jeffreys (1939), who described himself as a seismologist, discussed experiments in which the variable to be chosen is the distance of a seismic sensor from an explosion when usually the model to be fitted is simple linear regression in x. However, there are occasions when a higher-order polynomial is required and the experiments should at least allow a check of this. He also argued that, in field trials, there may be correlations between errors or underlying patterns of fertility, perhaps polynomials. How, then, should a design be chosen? Answers to these perceptive questions form an appreciable part of the literature to be reviewed.

The desire to find designs which are insensitive to omitted terms in the model is behind the development by Box & Draper (1963) of response surface methods, described in § 12·2.

The desire for balance against covariance leads to designs such as those of Williams (1952), and the presence of trends suggests systematic designs (Cox, 1951); see §§ 10·4 and 10·2, respectively.

In the next section we consider the three corner-stones of the Fisherian approach to experimental design. This is followed by sections on randomisation and factorial design, both of which were well developed in the first half of the 20th century. Section 6 describes the fundamentals of optimum design theory, since the results are applicable in the comparison of many kinds of design. We then consider for several sections what we have called agricultural design. In the latter part of the paper we return to industrial experiments and conclude with sequential experiments.

Since this paper reflects the topics covered in *Biometrika*, some subjects are treated in greater depth than others. Two pillars of the journal's work have been statistical problems arising in agriculture and in medicine, in both of which treatments are typically qualitative. Designs in these areas are more fully covered than are industrial designs. In the 1930s this work, following Fisherian principles, tended to be published in the *Supplement to the Journal of the Royal Statistical Society*. Examples are Tippett (1935) on problems arising in the cotton industry and Daniels (1938) on those in the wool industry. Industrial designs with quantitative factors are poorly represented in *Biometrika*. The work of Kiefer and of Box following the Second World War has already been mentioned and is covered to some extent, but there is virtually nothing about designs for off-line process control, described for example in Logothetis & Wynn (1989), whereby Western countries rejuvenated their manufacturing industry by adopting Japanese ideas on industrial experimentation. An exception is the founding paper of Plackett & Burman (1946).

We have found some subjects and some papers hard to categorise. One paper is Cheng & Li (1987), which explores the relationship between optimum experimental design and survey sampling. The relationship between the two fields seems to be strong and is explored and used elsewhere by Giovagnoli & Martino (2001) in designing epidemiological sampling schemes. A second paper that we found hard to fit into our scheme is Cox (1957), which considers designs when there is a preliminary measurement on each unit. This raises questions of randomisation and blocking which are important in clinical trials; see § 13. Since David Cox was editor of *Biometrika* for much of the period under review, we want to record the importance of his work on the design of experiments, especially his steadily-selling book Cox (1958b). The review by Mallows (1959) is so contorted that little of the book's value is conveyed.

Half a dozen influential books on the design of experiments were published in the 1950s, all reviewed in *Biometrika*. Apart from Cox's book, the reviewers' verdicts have stood the test of time. Johnson (1951, 1958) and Moser (1953) noted that Cochran & Cox (1957), whose first edition appeared in 1951, is a compendium, or manual, of the more useful designs: keep it on your shelf and use it if you want to copy out, for example, a Latin square or incomplete-block design. Johnson (1951) contrasted this with Mann (1949), whose mathematical theory enables you to construct such a design for yourself. Pearce (1956) noted approvingly that Finney (1955) would explain why you should use such a design. As Moser (1953) remarked, Kempthorne (1952) gives admirable treatments of both randomisation and factorial designs. McMullen (1955) pointed out that the previous books are biased towards agricultural applications: for industrial applications see Davies (1956), the first edition of which appeared in 1954.

Other books from the period have not lasted as well as their competitors, and the reviews show why. Cox (1953) observed that Quenouille (1953) was less satisfactory than

other books on both practical and theoretical matters. Pearce (1956) contrasted Federer (1956) unfavourably with Finney (1955) because it is divorced from practice. Chew (1958), reviewed by Johnson (1959), has interesting papers on design for industry but is patchy overall. An excellent recent introductory text, written in a non-technical style, is Cobb (1998).

3. Fisherian fundamentals

R. A. Fisher did more than anyone else to establish the theory and practice of designs for experiments of the 'agricultural' kind. He did this while he was at Rothamsted Experimental Station (1919–1933), where he was motivated by real agricultural experiments. Unfortunately, none of this was reported in *Biometrika*.

Some early correspondence between Fisher and Karl Pearson, then editor of *Biometrika*, was reported by E. S. Pearson (1968). In 1916 Fisher praised *Biometrika* warmly; a few years later Pearson rejected two of his papers because they were too controversial. According to Fisher Box (1978, p. 83), it is said that Fisher 'vowed never again to submit a paper to Pearson's *Biometrika*, and he never did'. This is confirmed by Reid (1998, p. 56). He concentrated on his books (Fisher, 1925, 1935), and published in other journals. His bitterness towards Pearson the man, combined with respect for his 'great enterprise in publication', still showed strongly in Fisher (1956), 20 years after Pearson's death.

Fisher is traditionally credited with emphasising, if not introducing, 'replication, randomisation and local control'. To these should be added the initial articulation of both treatment structure and block structure.

Sufficient replication is needed to distinguish genuine treatment differences from experimental error. If there is no structure to either the treatments or the experimental units, the design problem is simply that of choosing sufficient replication. Once the Neyman–Pearson significance-testing framework had been established, the question came to be phrased as follows: if we have two treatments, what replication is needed in order to have a given power β of detecting a given magnitude in the standardised difference between the treatment effects when a given significance level α is used? The difference is standardised relative to the standard deviation, whose value must be guessed; and responses are usually assumed to be normal. In clinical trials this is probably still the most frequently asked design question even today, in spite of developments in analysis for binary responses and for censored survival times. For two treatments and no block structure the answer can readily be obtained from t tables. For more treatments the process is more complicated, because the noncentral F distribution is not a simple shift of the usual F distribution. *Biometrika* has always assumed a responsibility for publishing statistical tables and charts. Pearson & Hartley (1951) gave charts which can be used iteratively, given the noncentrality parameter: replication determines the number of degrees of freedom for error, which determines power. However, the noncentrality parameter is not a very easy concept to non-specialists when there are more than two treatments. Kastenbaum et al. (1970a, b) and Bowman (1972) gave tables which use instead the standardised maximum difference between treatment effects.

Fisher insisted that every design must be randomised, although he gave different reasons for this in different places; see § 4. If there are no blocks then the design is determined by the replication of each treatment, giving a total of, say, n experimental units altogether. In a completely randomised design the treatments, with their given replications, are first

assigned to the experimental units systematically, and then a permutation is chosen at random from the $n!$ permutations of the experimental units.

'Local control' means the recognition that not all experimental units are alike, and the consequent design of the experiment to allow for this heterogeneity. A group of similar experimental units is called a block. In some applications the block boundaries are inherent: for example, when experimental units are half-leaves then the whole leaves are blocks. On the other hand, in a field trial there are probably continuous fertility gradients and the experimenter has some freedom to decide how large a block should be. In Fisher's randomised complete-block designs, each block has as many experimental units as treatments; randomisation is performed separately and independently within each block. Incomplete-block designs were introduced later; see § 7. Only in the 1980s did British statisticians, with a few earlier exceptions, begin to make a serious attempt at allowing for continuous fertility gradients in the design and analysis of field trials; see § 10.

Sometimes it is necessary to have more than one system of blocks. Using the rows and columns of a rectangular array as two sorts of block gives an approximate allowance for a fertility gradient in two dimensions. One special design for this structure is the Latin square, which has the same number of treatments, rows and columns: each treatment occurs once in each row and once in each column. Since the efficacy of a randomisation procedure depends on the purpose of randomisation, and Fisher was apparently so ambivalent about the latter, we find it curious that he was so sure that the correct randomisation procedure for a Latin square was random choice from among all Latin squares of that size; see his letter to MacMahon, cited in Bennett (1990, p. 273). This led to the hard combinatorial problem of finding all Latin squares of given size n. Although this has still not been solved for $n > 10$, Jacobson & Matthews (1996) gave an ergodic Markov chain whose state distribution converges to the uniform distribution on all Latin squares of given size.

The rows and columns of a Latin square are said to be crossed. At the other extreme, two systems of blocks may be nested: each of the larger type consists of several of the smaller. The early work at Rothamsted included designs of this type where the smaller blocks are called whole plots. They are necessary for experiments where the treatments consist of all combinations of the levels of two treatment factors, one of which can be applied only to relatively large areas. The levels of this factor are applied to the whole plots in a randomised complete-block design; then each whole plot is split into subplots, to which levels of the second factor are applied. This is called a split-plot design.

When the treatments consist of all combinations of the levels of two or more treatment factors, the treatment structure is said to be factorial. Fisher (1935) was eloquent about the advantages of a factorial experiment over a sequence of experiments in which only one factor is changed at a time. Suppose that A and B are two factors, with n and m levels respectively. If the responses on all combinations of those levels may be arbitrarily different then nm parameters are needed to describe them. However, if the effect of changing the level of A is the same at each level of B then only $n + m - 1$ parameters are needed and the response is said to be additive in A and B. The interaction between A and B is defined to be the departure of the response from the additive model. This can be generalised to several factors: the interaction between s factors covers everything that cannot be accounted for by interactions between $s - 1$ or fewer of those factors. The art of designing factorial experiments is in creating designs that are good for estimating those main effects and interactions deemed important by the experimenter under the assumption that certain other interactions are zero; see § 5.

The split-plot experiment is the first example where two different error variances appear in the model and the analysis. Suppose that A is applied to whole plots and B to subplots. Then the main effect of A is assessed against the variance of whole plots within blocks, while both the main effect of B and the A-by-B interaction are assessed against the variance of subplots within whole plots.

Fisher and F. Yates, who joined him at Rothamsted in 1930, devised increasingly complicated one-off experiments for factorial treatment structures in many systems of blocks, with correspondingly complicated tables for the analysis of variance. These can be seen in the sequence of Rothamsted Annual Reports. The general constructions for factors with two or three levels were described by Yates (1935, 1937). The story is continued in §§ 5 and 9. Putting these designs into a general framework is an ongoing task.

4. Randomisation

Randomisation is a much misunderstood topic, but those associated with *Biometrika* over the century can be proud of the journal's contribution to discussion of the issues. There are so many different opinions about what randomisation is for that it is hard even to pose the right questions, let alone answer them satisfactorily.

The most basic purpose of randomisation is to remove bias from the estimation of treatment effects. This can be achieved by taking a systematic plan such as

$$ABABABAB\ldots \quad \text{or} \quad ABBAABBA\ldots$$

and randomly allocating treatments to the letters. Even today, some experimenters do this. Efron (1971) explained why this will not do. If experimental subjects enter the trial sequentially then the experimenter can be influenced by knowledge of the next treatment in selecting the next subject; see § 13 for more on sequential randomisation. Even if all experimental units are allocated to treatments at the same time, such simple randomisation is no protection against what Efron called 'accidental bias', i.e. systematic trends which the experimenter has not suspected.

Theoretical discussion of randomisation often overlooks the fact, verified by our own practical experience, that it offers the only protection against a deliberate but unmalicious bias introduced by some experimenters by balancing over certain block factors without declaring them; here the bias occurs in the estimate of error. Yates (1939) phrased this idea slightly differently: 'randomization provides an assurance ... to [the] sceptical ... that the magnitude of the ordinary sources of disturbance ... has been evaluated by means of the estimate of error'. From a different perspective, Stone (1969) came to a similar conclusion: although Bayesians may not appear to need randomisation, their results will be more credible to other statisticians if they do randomise.

A magnificent series of papers in *Biometrika* in the late 1930s debated the purposes of randomisation and whether or not these are achieved in the simple randomised designs. Welch (1937), Pearson (1937) and Pitman (1938) believed that the purpose of randomisation is to enable the data analyst to do a randomisation test: apply all possible outcomes of the randomisation to the same data, calculate the test statistic in each case, and see where the observed test statistic lies in the distribution. This point of view was clearly expressed by Kempthorne (1952, Ch. 7), Kempthorne & Doerfler (1969) and Efron (1971). The confusion surrounding the whole topic is evident from the comment in Kempthorne (1975) that his paper with Doerfler had been rejected many years earlier by the *Journal*

of the American Statistical Association because it was both wrong and so obvious that it was well known.

A related view is that randomisation helps to justify the assumption of normal errors; see also Collier & Baker (1963, 1966). These approaches become untenable as soon as the block structure becomes complicated. The foregoing authors deem a randomisation procedure satisfactory if the upper tails of the randomisation distribution are close to those predicted by normal theory when there are no treatment effects; in other words, their interest is in the significance level of the test.

"Student" (1938) took a different approach, in a public disagreement with Fisher after many years of friendly correspondence. Gosset wanted to 'balance' for possible trends, which will tend to reduce the bias and the variance of the estimators of treatment differences. However, as both men showed, if the trend is not fitted then error is overestimated. Fisher's solution was to randomise, Gosset's to balance. Unlike the previous batch of authors, who had tested their ideas by applying randomisation tests to uniformity data, Gosset added treatment constants to uniformity data and then did analysis of variance, thus assessing the power of the normal-theory test. Neyman & Pearson (1938) and Pearson (1938) provided further explanation in support of Gosset's arguments.

The point of view of Yates (1939) was that the purpose of experiments is not testing but estimation. Apart from the avoidance of accidental bias, the most important purpose of randomisation for him was to obtain unbiased estimators of error, and hence of variances of treatment differences. He called a randomisation procedure valid if it has this property. Anscombe (1948), Johnson (1948), Nelder (1954) and Bailey (1981) took this approach a step further, by arguing that randomisation justifies the assumption not of normal errors but of certain patterns in the first and second moments of the errors.

This approach is consistent with that of Kempthorne (1952, Ch. 8) and his school, who argue that the response is additive in treatment effect and arbitrary plot effect, and hence that the randomisation procedure determines which mean squares should be used to estimate which error variances. A fine paper in this tradition was Folks & Kempthorne (1960), which showed how to use the data from an incomplete-block experiment to estimate what the various mean squares would have been had the experiment been differently blocked. This followed similar work of Yates (1935) for simpler designs.

Although within this school, Wilk (1955) did not assume additivity between treatments and plots. Nor did Cox (1958a), who refuted Wilk. In these circumstances it is hard to interpret what a treatment effect means, as Cox discussed.

Jeffreys (1939) reviewed the arguments of the 1930s and concluded that you should 'balance or eliminate the larger systematic effects as accurately as possible and randomise the rest'.

It is astonishing how much Gosset's and Yates's opinions on the purposes of randomisation have been ignored by later writers, particularly those like Efron (1971) who deal with sequential randomisation. The papers by Wei (1978) and Begg (1990) and the ensuing discussion show how much confusion there is between the idea of randomisation as the justifier of a simple model and randomisation as the justifier of a randomisation test.

Turning to somewhat complicated block structures, Yates (1939) noted that, although his approach works for Latin squares and split plots, there is no valid randomisation for the 'Latin square with balanced corners'. Preece et al. (1973) observed the same thing for the three-way cross with no interactions. A simple specification of the block structure does not necessarily go hand in hand with a clear story about randomisation.

On the other hand, Yates's approach does have two definite advantages. If you are not

trying to approximate the normal distribution, or to perform a randomisation test with given power, then there is no need to seek a randomisation procedure with a large number of possible outcomes. For example, if a Latin square is needed, validity is ensured by taking any one square and randomising its rows and columns; see Anscombe (1948). Equally, it may be possible to restrict the usual randomisation so as to avoid some undesirable plans while still retaining validity. Some examples are given by Bailey (1983, 1987).

5. Factorial experiments

Let $A_1, \ldots, A_m$ be treatment factors, and suppose that A_i has n_i levels. When all of the n_i are equal, the factorial structure is said to be symmetric. Fisher (1942, 1945) devised a way of arranging symmetric factorial experiments in blocks when the common number of levels is a prime p. The method uses groups, in the algebraic sense; it systematises and generalises the methods of Yates (1935, 1937).

Denote the levels of each factor by the integers modulo p. The treatment which has factor A_i at level s_i for $i = 1, \ldots, m$ is written $a_1^{s_1} a_2^{s_2} \ldots a_m^{s_m}$. Under the rule that $a_i^p = 1$, the treatments form an Abelian group G. Define formal words $A_1^{r_1} \ldots A_m^{r_m}$, for $0 \leqslant r_i \leqslant p-1$, as functions on the treatments by the rule

$$A_1^{r_1} \ldots A_m^{r_m}(a_1^{s_1} \ldots a_m^{s_m}) = \sum_{i=1}^{m} r_i s_i \bmod p. \tag{1}$$

Then each formal word partitions the treatments into p sets according to the value of (1). The $p-1$ degrees of freedom for the differences between these sets all belong to the interaction between those factors A_i for which $r_i \neq 0$.

The set of formal words also forms a group G^*, under the rule that $A_i^p = 1$. Let H be any subgroup of G^*, and define H^0 to consist of those treatments for which (1) is zero for all formal words in H. Then H^0 is a subgroup of G. Construct the design by allocating each coset of H^0 to a block. Fisher showed that every effect outside H is estimable; those words in H are said to be confounded with blocks.

Two important papers in 1947 recast Fisher's method in terms of subspaces of an affine space over the finite field GF(p). Kempthorne at Rothamsted did this in *Biometrika* (Kempthorne, 1947). Independently, R. C. Bose was working in Calcutta. Although he published very little in *Biometrika*, his statistical education had consisted of reading the 1900–1932 issues of 'the most important statistical journal at that time' and Fisher's (1925) first book; see Bose (1982). Impressed by Fisher's talks on his visit to India in the late 1930s and also much influenced by the expatriate German geometer F. W. Levi, Bose developed his ideas on confounding in Bose & Kishen (1940) and gave a more general version in Bose (1947). Of course, a finite field with p elements exists if and only if p is a prime power, and a myth grew up that there is no systematic way of constructing factorial designs when p is not a prime power, or any asymmetrical factorial design.

This is odd, because Fisher & Yates (1938) had already laid the foundation for the general case in their work on cyclic incomplete-block designs. Let K be any subset of a group G, and g any element of G. Then the translate of K by g is defined to be $\{kg : k \in K\}$: if K is a subgroup then its translates are called cosets. In a cyclic incomplete-block design the blocks are all the distinct translates of one or more subsets of a cyclic group G; see § 7. Fisher's factorial designs are the same except that G is elementary Abelian rather than cyclic.

It took over a quarter of a century before John (1973) combined the two methods to produce general factorial designs which he called 'generalized cyclic'. The treatments are $a_1^{s_1} \ldots a_m^{s_m}$, where $0 \leqslant s_i \leqslant n_i$. They form a group under the rule $a_i^{n_i} = 1$. The blocks are now all the translates of one or more subsets of G.

There remained the problem of identifying totally or partially confounded effects in such designs. This was solved by Bailey et al. (1977) and Bailey (1977), who generalised (1). If $m = 3$, $n_1 = 6$, $n_2 = 2$ and $n_3 = 3$ then (1) is replaced by

$$A_1^{r_1} A_2^{r_2} A_3^{r_3}(a_1^{s_1} a_2^{s_2} a_3^{s_3}) = r_1 s_1 + 3r_2 s_2 + 2r_3 s_3 \bmod 6. \tag{2}$$

The powers of, say, $A_1 A_2 A_3^2$ are $A_1 A_2 A_3^2$, $A_1^2 A_3$, $A_1^3 A_2$, $A_1^4 A_3^2$, $A_1^5 A_2 A_3$ and I, so the five degrees of freedom associated with this formal word consist of two degrees of freedom for the A_1-by-A_2-by-A_3 interaction, two degrees of freedom for the A_1-by-A_3 interaction and one degree of freedom for the A_1-by-A_2 interaction.

For Fisher's original construction, the words in H should be long. Finney (1945) extended the work to fractional replicates, in which a single block is the whole design. If all words in H have at least $2d + 1$ letters then all interactions between d or fewer factors can be estimated if we can assume that all interactions between $d + 1$ or more factors are zero. Such a fraction is now said to have resolution $2d + 1$.

Two papers in *Biometrika* in the 1940s used the group theory method. Plackett (1946) constructed symmetric fractions with high resolution, while Brownlee & Loraine (1948) found sets of mutually orthogonal Latin squares and their analogues in higher dimensions.

Coding theory appeared in the late 1940s and early 1950s (Hamming, 1950). For a good fractional factorial formed by the Fisher–Finney method, the subgroup H is a good linear code over GF(p): a fraction of resolution $2d + 1$ gives a code that can correct up to d errors. So-called saturated fractions of resolution 3 are the same as, or rather dual to, Hamming codes. Some results in coding theory now usually attributed to Hamming were originally given by Fisher (1942, 1945).

The first statistical paper to mention word length appears to be Brownlee et al. (1948). This had little immediate influence, but has led in the longer term to such concepts as resolution (Box & Hunter, 1961) and minimum aberration (Fries & Hunter, 1980), which have largely been developed in the pages of *Technometrics*. A recent paper in *Biometrika* is Fang & Mukerjee (2000). It also foreshadowed the concept of weight in coding theory. Some of the *Technometrics* papers give remarkably little clue about the original development of fractional factorials; in particular, Box & Hunter (1961) completely ignored the table of designs given in *Biometrika* by Mitton & Morgan (1959).

The group method is not the only way of constructing factorial designs. In a landmark paper, Plackett & Burman (1946) introduced another form of unblocked design containing only a fraction of the complete set of factorial treatments. Their so-called main-effects plans, which are constructed from Hadamard matrices, have resolution 3: for every pair of treatment factors, all combinations of their levels occur equally often.

At this point, *Biometrika* missed a chance to be extraordinarily influential. In the foreword to Hedayat et al. (1999), C. R. Rao writes:

> Since my paper provided a generalization of the multifactorial designs of Plackett and Burman (1946) which appeared in *Biometrika*, I submitted it to the same journal for publication. I was disappointed when I received a letter from the editor, E. S. Pearson, stating that the paper was too mathematical for *Biometrika* and the applications discussed were not significant enough for publication. I decided to split the paper into two parts. The part dealing with the general theory of orthogonal arrays I sent to the *Proceedings*

of the Edinburgh Mathematical Society. The editor commented on the paper as "highly original" and published it (Rao, 1949). The part dealing with applications was sent to the *Journal of the Royal Statistical Society* where it was accepted without any revision and published (Rao, 1947).

Orthogonal arrays of strength t are exactly the same as certain fractional factorial designs with resolution $t+1$. Today, orthogonal arrays are widely used in factorial design, both for $t=2$ and for higher values of t. For $t=2$ they have many other uses in design and sampling, which can all be seen in *Biometrika*: in the construction of optimal resolvable incomplete-block designs (Bailey, Monod & Morgan, 1995); in valid procedures for restricted randomisation (Bailey, 1987); in estimating variance components (Gupta & Nigam, 1987) and nonlinear statistics (Wu, 1991) from samples; in optimal designs for observations which are spatially correlated within blocks (Martin & Eccleston, 1991, 1998); and even in designs for continuous factors (Covey-Crump & Silvey, 1970).

Two-level fractional factorials are sometimes used in screening experiments on many quantitative factors, of which it is hoped that only a few are effective. A response surface design will then be performed on these few; see § 12·2. Fedorov et al. (1968) described an experiment in which three factors were chosen from ten after a screening experiment on 16 units; a more detailed experiment was then performed on those three factors, using more than two levels. Usually the first stage is done using an orthogonal array, although Fedorov et al. (1968) did not. Orthogonal arrays with the same strength can be compared by using their projective properties; see Box & Tyssedal (1996), Cheng (1998) and Tsai et al. (2000). Of course, this is heavily dependent on correct assumptions about zero interactions, so Cotter (1979) and Elster & Neumaier (1995) introduced fractions of a different type: each main effect can be assessed at one or two particular settings of all the other factors, but the designs do not seem to be as good for their size as fractions with resolution 4. At the other extreme are so-called supersaturated fractions, with more factors than experimental units. These have not been much addressed in *Biometrika*, but one example is by Wu (1993).

6. Optimum design for regression

6·1. *General theory*

The ideas behind optimum design are set out by Kiefer (1959) in a paper read to the Royal Statistical Society. The published discussion suggests that the paper had a rough reception, although the memory of those present is that the meeting itself was amicable. The hostility that Kiefer felt arose from the written comments. The starting points for Kiefer's work were the suggestion of Wald (1943) to compare designs using D-optimality, and the work of Elfving (1952) on two-variable regression problems. Although many of Kiefer's ideas were formed by 1959, a major subsequent development was the various forms of the general equivalence theorem, originally relating G- and D-optimality (Kiefer & Wolfowitz, 1960). A short history of the development of Kiefer's ideas is given by Wynn (1985) in the introduction to Kiefer's collected papers on design. Kiefer (1959) surveys earlier history including the work of Yates and mathematical approaches to weighing designs.

An influential early book on optimum design is Fedorov (1972), which has been followed by several books on regression design, including Silvey (1980), Pazman (1986), Pukelsheim (1993), Fedorov & Hackl (1997) and, most practically, Atkinson & Donev (1992). Shah & Sinha (1989) focus on designs for discrete treatments.

The central theory is most conveniently described for models which may be linear or nonlinear in the parameters. For univariate nonlinear models

$$y_i = \eta(x_i, \theta) + \varepsilon_i \quad (i = 1, \ldots, n), \tag{3}$$

the exact design problem is to obtain an n-point design to estimate some function of the p-dimensional parameter vector θ with high efficiency. In (3) the errors ε_i are independently and identically, and perhaps normally, distributed, and x_i is the vector of m known explanatory variables for the ith experiment. One of Kiefer's important contributions was to replace the exact design problem with the approximate problem in which the designs are of the form

$$\xi = \begin{Bmatrix} x_1, x_2, \ldots, x_r \\ \omega_1, \omega_2, \ldots, \omega_r \end{Bmatrix}, \tag{4}$$

where the r design points x_i are distinct elements of the design space $\mathscr{X}$ and the associated weights ω_i are positive and sum to one. The conversion of optimum approximate designs to near-optimum exact ones by setting $\omega_i = n_i/n$ for integer n_i is discussed by Pukelsheim & Rieder (1992).

If the variance of the errors ε_i is taken as one, the information of a design ξ is

$$M(\xi, \theta) = \sum_{i=1}^{r} \omega_i \frac{\partial \eta(x_i, \theta)}{\partial \theta} \frac{\partial \eta(x_i, \theta)}{\partial \theta^{\mathrm{T}}} = F^{\mathrm{T}} \Omega F, \tag{5}$$

say, where Ω is a diagonal matrix with ith diagonal element ω_i. For models linear in the parameters, F does not depend on θ and the information is written $M(\xi)$, with $f(x_i)$ the ith row of F.

Optimum experimental designs typically minimise some convex function of the inverse of the information matrix. Three of the criteria can be interpreted, for linear models, both in terms of the variance of parameter estimators and as functions of the eigenvalues λ_i of $M(\xi)$. A fourth criterion is related to the first.

1. *D-optimality* minimises $-\log|M(\xi)|$, or equivalently maximises $|M(\xi)|$. The design therefore minimises the generalised variance of the parameter estimators, equivalent to the volume of a confidence region for the parameters. Equivalently, the product of the eigenvalues of $M^{-1}(\xi)$,

$$\prod_{i=1}^{p} 1/\lambda_i,$$

is minimised.

2. *A-optimality* minimises the average variance of the parameter estimators, equivalent to minimising

$$\sum_{i=1}^{p} 1/\lambda_i.$$

3. *E-optimality* minimises the variance of the least well-estimated contrast $a^{\mathrm{T}}\theta$ with $a^{\mathrm{T}}a = 1$, equivalent to minimising $\max(1/\lambda_i)$.
4. *G-optimality*. The variance of prediction at x is

$$d(x, \xi) = f^{\mathrm{T}}(x) M^{-1}(\xi) f(x), \tag{6}$$

since σ^2 is taken as one. In G-optimum designs the maximum value over the design region of $d(x, \xi)$ is minimised.

For models in which the information matrix depends on θ the results hold for locally optimum designs based on an initial parameter value θ^0. Thus locally D-optimum designs minimise $-\log|M(\xi, \theta^0)|$ and so minimise the volume of the asymptotic confidence region for the parameters. Examples are given in § 12·5. There is usually no dependence of the design on parameter values for linear models, except if interest is in estimating a nonlinear function of the parameters; see § 12·6.

A major theoretical development is the series of equivalence theorems. In the first, Kiefer & Wolfowitz (1960) prove that if a design ξ^* is D-optimum it is also G-optimum, and vice versa. Thus criteria 1 and 4 above yield the same optimum design. Such equivalence theorems both suggest algorithms for the construction of designs and provide methods for checking the optimality of any continuous design.

Optimum designs in general depend on the optimality criterion; if they are optimal for a family of convex criteria based on the information matrix, they may satisfy the conditions for universal optimality (Kiefer, 1975b). Such designs are rare, being identifiable only for discrete factors; one, for example, is a Latin square. References are in Yeh (1986). Regression designs depend also on the model, the design region $\mathscr{X}$ and the number of trials. A useful example is the second-order response surface model in two factors,

$$E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2, \tag{7}$$

with design region $\mathscr{X}$ the unit square $-1 \leqslant x_1, x_2 \leqslant 1$. The continuous D-optimum design is supported on the points of the 3^2 factorial. These points can be divided into sets having 0, 1 and 2 nonzero coordinates and so are the centre point of $\mathscr{X}$, the four centres of sides and the corners of the region, which are the points of the 2^2 factorial. The continuous D-optimum design puts different weights on these three sets. A good integer approximation can be found for $n = 13$ by repeating the corner points of $\mathscr{X}$, but the design does not satisfy the equivalence theorem.

A measure of the advance made by Kiefer can be obtained by comparing his approach with that of Tocher (1952b), who was the proposer of the vote of thanks for Kiefer's read paper. Tocher, in the Introduction to his paper on block designs, eloquently articulates the need for a general theory of design and looks for designs in which treatment contrasts have the same variance. Unfortunately he was unaware of the unification arising from the result on D-optimality in Wald (1943), whose example was Latin and Graeco-Latin squares. In Tocher's note in *Biometrika* (Tocher, 1952a), as in much of the earlier work, the emphasis is on orthogonality, which is difficult to extend to second-order models such as (7).

6·2. *Specific theoretical problems*

After 1959 the work of Kiefer and his collaborators continued to be published in American journals and books, for example Farrell et al. (1967). Within 10–12 years there were, in addition, active research groups in optimum design in Glasgow, Moscow and at Imperial College, London. References to this period of work are given in Kiefer (1974). In 1974 a conference on optimum design with Kiefer as a speaker was organised at Imperial College by David Cox, who was then editor of *Biometrika*, and this led to four papers in this journal.

Kiefer's published contribution to the conference (Kiefer, 1975a) was to investigate how optimum designs changed with smooth changes in the design criterion. The D-, A- and E-criteria were described in the previous section in terms of the eigenvalues λ_i of $M(\xi)$.

They can be combined in the function

$$\Phi_k(\xi) = \left\{ p^{-1} \sum_{i=1}^{p} \lambda_i^{-k} \right\}^{1/k}, \tag{8}$$

which gives D-, A- and E-optimality for $k = 0$, 1 and ∞. The example in the paper is quadratic regression on the simplex. The properties of the designs are compared with the response surface designs to be described in § 12·2. The remaining papers from the conference are here considered according to subject. An extension of Kiefer's work is given by Dette & O'Brien (1999), whose smooth family is a power function of the variance $d(x, \xi)$. Thus average variance is included as well as G-optimality.

All design criteria described so far are based on the information for all parameters in the model. Often only a subset of s parameters is of interest, when, for example, D-optimality is replaced by D_s-optimality in which information about the subset of parameters is maximised. Several papers consider the aspects of D_s-optimality or related problems.

Laycock & Silvey (1968) look at designs for estimation of the highest-order term in a one-factor polynomial. In the discussion of Wynn (1972) both Sibson and Silvey show that a dual to the optimum design problem is to find an ellipsoid of smallest volume, centred at the origin, which encloses the image of the design region generated by the terms in the model. The corresponding problem for D_s-optimality is solved by Silvey & Titterington (1973) and requires a cylinder of minimum volume. The extension to general design criteria is in Silvey & Titterington (1974). Titterington (1975) considers the reverse problem of interpreting problems involving minimum-volume ellipsoids with arbitrary centres as design criteria.

A technical difficulty in proving theorems about D_s-optimality is that some of the optimum design matrices are singular, as they are for some other criteria. Such problems are considered by Silvey (1978). A useful general reference on equivalence theory is Whittle (1973).

The equivalence theorem for D-optimality implies that designs for estimation of parameters also have good properties for the variance of prediction over the design region. Designs for prediction are concerned with estimating the response outside this region. Dette & Wong (1996) find designs for extrapolation of polynomial models in a single variable, when the order of the true model is uncertain. The earlier paper of Herzberg & Cox (1972) assumes the multifactor model is known.

Laycock (1975) investigates designs when the response or the explanatory variables are directions. A specific problem involving directions is the migration of fronts of groups of lobsters. The design problem is the location of lobster traps. The D-optimum design of Yang (1976) puts all traps on the boundary of the region of the sea-bed being monitored. This result is typical of D-optimality in spatial problems (Müller, 2000).

6·3. *Bayesian designs and previous observations*

Some prior information is available before most experiments are designed, but has not so far been formally included in our design criteria. We now consider the incorporation of prior information, which may be of beliefs about the parameters of the model, or, as in the paper of Covey-Crump & Silvey (1970), observations from an earlier experiment. They find exact D-optimum designs when the variance of the observations is assumed known. Guttman (1971) shows that the same designs are obtained if the variance is not

known. Verdinelli (2000) establishes conditions on the design criterion for Bayesian designs to be independent of the prior distribution of σ^2. Dette & Wong (1998) find D-optimum designs for polynomial models in which the variance is not constant but has an exponential form with unknown parameters.

The use of hierarchical linear models provides a tractable method for introducing prior information into experimental design. Smith & Verdinelli (1980) use such priors for linear models to find designs for several simple examples including the one-way layout and dose–response curves. They find that, if the prior assumptions hold, their designs are an improvement on standard D-optimum designs. Giovagnoli & Verdinelli (1983) find Bayesian A-, D- and E-optimum designs for the two-way layout of blocks and treatments when one of the treatments acts as a control.

The preceding designs are all for fixed effects. The hierarchical prior can be thought of as generating the particular parameter values encountered, but interest remains in the particular values. Lohr (1995) finds designs for the random effects one-way layout, with interest in functions of the variances of the two strata. Non-Bayesian designs for variance components are given in *Biometrika* by Mukerjee & Huda (1988), and elsewhere by Bogacka (1995).

A third class of design problems for which Bayesian solutions have been found have to do with prediction. Eaton et al. (1996) use a very general framework to obtain designs compared by the predictive distributions they yield. Examples include linear regression in several factors, the one-way layout and logistic regression. The earlier papers of Brooks use a decision-theoretic approach, Brooks (1974) to prediction and Brooks (1977) to design for the purpose of controlling the response at a specified value.

6·4. *Discrimination between models*

Suppose there are two competing regression models, neither of which is a special case of the other and one of which is true. One way to proceed is to combine the two models into a hypermodel and then to use D_s-optimum designs to provide good estimates of the parameters distinguishing the models. Unless the models differ by only a single parameter, more powerful tests are obtained by performing experiments where the two models are furthest apart. Since the parameters of the models have to be estimated, the point at which the models are furthest apart will depend on the design. Atkinson & Fedorov (1975a) introduced T-optimum designs which maximise the power of the F test for departures from the false model in the direction of the true model. The designs are called T-optimum because one of the authors felt that it might be considered immodest to call them F-optimum.

The extension to designs for more than two models is described by Atkinson & Fedorov (1975b). These two papers bring discrimination between models within the framework of convex design theory. Fedorov & Khabarov (1986) continue the exploration of convex design theory for model discrimination, establishing a relationship between designs for parameter estimation and a degenerate form of T-optimality. Hill (1978) finds the same D-optimum design for three of the models in an example of Atkinson & Fedorov (1975b). Such a structure makes it easy to find designs which are efficient for both model discrimination and parameter estimation.

Jones & Mitchell (1978) consider model inadequacy in which one of the two models is a special case of the other. Provided the models differ by more than one parameter, the T-optimum design depends on the value of the vector of extra parameters. The authors

derive criteria based on properties of the noncentrality parameter for the test of the smaller model and compare the results with T- and D_s-optimality. Ponce de Leon & Atkinson (1991) include prior information in T-optimality and give equivalence theorems when there is information both on the parameters of the models and on the probability that each is true.

7. INCOMPLETE-BLOCK DESIGNS

Consider a block design for v treatments in b blocks. For simplicity of exposition here we assume that each treatment has replication r and each block has size k, although the theory has now developed to include unequal replication and unequal block size. Let N be the treatments by blocks matrix whose (i, l)th entry N_{il} is the number of times treatment i occurs in block l. The design is called binary if every N_{il} is in $\{0, 1\}$. Also, NN^{T} is called the concurrence matrix: its (i, j)th entry is the concurrence of treatments i and j. The information matrix L is defined by $L = rI - k^{-1}NN^{\mathrm{T}}$. The variance of the estimator of the difference between the effects of treatments i and j is proportional to $L_{ii}^{-} + L_{jj}^{-} - 2L_{ij}^{-}$, where L^{-} is a generalised inverse of L.

There has been some confusion in the literature between ways of constructing block designs, their combinatorial properties and their statistical properties. Important construction methods include: (i) identifying the treatments with a crossed factorial structure and using confounding, which gives, for example, the square and cubic lattices of Yates (1936a); (ii) the cyclic and generalised cyclic constructions described in § 5; (iii) projective geometry (Bose, 1939); (iv) building designs out of smaller designs (Kurkjian & Zelen, 1963) or from Latin squares or orthogonal arrays (Muller, 1965, 1966).

Patterson & Williams (1976) introduced what they called α-designs, which are very useful for variety trials when v is large but r is small. Jarrett & Hall (1978) generalised their method of construction as follows. Identify the treatments with $G \times T$, where G is a cyclic group and T is a set of size at least 2. For g, h in G and t in T, define $(g, t)h$ to be (gh, t). Then blocks are all translates of one or more initial blocks, just as in the cyclic construction. Unfortunately, the authors called their method 'generalized cyclic', although it is not at all the same as the generalised cyclic method described in § 5.

Combinatorial properties include resolvability, balance and partial balance. A design is called resolvable if its blocks can be grouped into sets which contain each treatment once. These designs are necessary in applications such as variety trials, where many regulatory authorities still insist that complete-block designs should be used. The importance of α-designs is that they are resolvable and available for all values of v and k for which k divides v.

A binary design is said to be balanced if all the concurrences are equal. In early work Yates (1936b) commented that he had been unable to find a balanced design with $v = 16$, $r = 3$, $b = 8$ and $k = 6$. Soon afterwards, Fisher (1940) proved his famous inequality $b \geqslant v$, which was generalised to unequal block sizes by Atiqullah (1961) and Raghavarao (1962). Thus balanced designs are often too large for real experiments.

Partial balance, introduced by Bose & Nair (1939), was largely ignored by *Biometrika*. The set of unordered pairs of treatments is partitioned into classes, which have to satisfy a technical condition so that they form what is called an association scheme. A design is partially balanced if all pairs in the same class have the same concurrence. Cyclic designs are partially balanced: the class containing $\{i, j\}$ depends on the value of $\pm(i - j) \bmod v$. So are square lattice designs and designs which are generalised cyclic in the first sense. A

factorial design has a special sort of partial balance, called factorial balance by Yates (1935), if the class containing i and j depends only on the set of factors which take the same level on i and j. Kurkjian & Zelen (1963) called this 'property A'.

There are two methods of doing the calculations to analyse data from an incomplete-block design. One involves finding L^-, which is easy if the design is balanced and relatively easy if it is partially balanced, because the technical condition ensures that the value of L_{ij}^- depends only on the class containing $\{i, j\}$. John (1965) noted this in *Biometrika* for designs with two classes, but it is true more generally (Bose & Mesner, 1959). Thus partially balanced designs are easy to analyse and the standard errors for differences can be presented simply, one for each class. If a design has factorial balance, every normalised contrast for each factorial effect has the same variance.

The second method was introduced in *Biometrika* by Stevens (1948). What is now called 'sweeping by blocks' is the calculation of block means, their addition to the working vector of block parameters and their subtraction from the working data vector. Alternately sweeping by blocks and treatments converges rather rapidly to the least squares fit, with no need for matrix inversion.

The most important statistical property of a block design is its efficiency. Pearce (1968, 1970, 1971) discussed this in a series of thoughtful papers. The efficiency for a single contrast is the ratio of two variances: that in a design with the same replication and no blocks, and that in the given design. It is the product of two quantities: the efficiency factor, which depends on the design and the contrast, and the ratio of the error variances in the two situations.

Three important *Biometrika* papers (Wilkinson, 1970; James & Wilkinson, 1971; Pearce et al., 1974) showed the crucial role of those contrasts which are eigenvectors of L. Pearce et al. (1974) called them 'basic contrasts'. Their efficiency factors are now called the 'canonical efficiency factors', whose harmonic mean is the A-criterion for optimality. Wilkinson (1970) showed that, if the basic contrasts, or even their eigenvalues, are known, then the method of sweeping can be modified to attain the exact answer in $m-1$ iterations, where m is the number of distinct eigenvalues of L. James & Wilkinson (1971) considered the vector space $\mathbb{R}^{vr}$ and its two subspaces consisting of vectors which are constant on treatments and blocks, respectively. This geometric viewpoint sheds a lot of light. The canonical efficiency factors are the squares of the sines of the critical angles between the two subspaces. The following generalisation of Fisher's inequality is immediately apparent: no matter whether the block sizes or replications are equal, if all canonical efficiency factors are less than unity then $b \geqslant v$. Surprisingly, Herr (1976) rediscovered James & Wilkinson's geometry without apparently realising that it was the same.

It is fairly obvious that, among binary designs, balanced designs are universally optimal, but this observation is of no use for designs with small numbers of blocks; see also Yeh (1986). There are few useful general results on A-optimality of unbalanced incomplete-block designs. Conniffe & Stone (1975) showed that certain partially balanced designs are A-optimal; their methods led to the more general results reported by Shah & Sinha (1989). *Biometrika* contains some results on E-optimality by Jacroux (1980), Gupta & Singh (1989) and Das & Kageyama (1991).

However, a two-pronged approach developed chiefly in *Biometrika* has been very useful in identifying designs which are near-optimal. Conniffe & Stone (1974), Jarrett (1977) and Paterson (1983b) gave upper bounds for the value of A; Patterson & Williams (1976), Cheng & Wu (1981), Hall & Jarrett (1981), Paterson (1983a) and Paterson & Wild (1986) gave heuristics for choosing the input to constructions such as cyclic designs, both mean-

ings of generalised cyclic and α-designs, whose value of A can then be compared with the upper bound.

8. Cross-over trials

When each experimental subject is used and measured in several time periods we have a repeated measures design or a cross-over trial. We prefer to use the former name for experiments in which each subject has the same treatment throughout, and the latter, or 'change-over trial', if each subject has the possibility of changing treatment each period. However, the names have been used interchangeably in the literature.

Jones & Kenward (1989) trace the early history of cross-over trials through crop-rotation experiments, feeding trials and bioassay. Their use in clinical trials and sensory experiments is relatively recent.

At its simplest, a cross-over trial is just a row–column design, with periods as rows and columns as subjects. So one might seek a good row–column design, for example, one in which each treatment occurs equally often in each row and equally often in each column. However, there is a danger that a treatment administered in one period may have a residual effect in the next period, and possibly second-, third-, ... order residual effects in subsequent periods.

A cross-over design is said to be balanced for first-order residual effects if each treatment follows every other treatment equally often. The simplest design of this type is the so-called column-complete Latin square, introduced by Williams (1949). The first comprehensive paper on more general cross-over trials balanced for first-order residual effects was by Patterson (1952). He also required that treatments be orthogonal to rows and balanced, in the sense of balanced incomplete-block design, with respect to columns, as well as satisfying some extra conditions related to the final period. His main method of construction used translates of ordered sequences in Abelian groups. It is interesting that this antedates the similar construction of factorial designs (John, 1973) by twenty years.

In clinical trials, animal feeding experiments and tasting trials, the number of periods is usually small, and it is often assumed that there are only first-order residual effects. Good designs for this situation were given by Davis & Hall (1969), Patterson (1973), Afsarinejad (1983) and Russell (1991), while Fletcher (1987) extended the ideas to factorial treatments. In a slight variant, the treatments are equally spaced levels of a quantitative factor, and it is assumed that only the linear contrast has a residual effect; see Berenblut (1968) and Patterson (1970).

In crop experiments, the period is a year; experiments may last for many years, and residual effects persist for a long time. Patterson (1968) introduced what he called 'serial' designs: if there are t treatments and t^N plots then every combination of direct and residual effects must occur once in every batch of N consecutive years; moreover, the confounding equation like (1) that gives the treatments in year i in terms of those in years $i-N$ to $i-1$ must be the same for all i. Serial designs with $N=2$ were further developed by Kok & Patterson (1976).

In the context of lactating cows, Brandt (1938) proposed that each cow might contribute not just a fixed effect but its own linear trend over time. Switchback designs were developed for this situation; more were given by Oman & Seiden (1988).

Berenblut & Webb (1974) suggested that the problem in cross-over trials might be not residual effects but correlated errors within each subject, with the magnitude of the correlation depending on distance apart in time. Their numerical investigations were backed

up by the theoretical work of Kunert (1985), who showed that column-complete Latin squares 'with balanced end pairs' are often optimal.

Matthews (1987) and Kunert (1991) assumed both correlated errors and first-order residual effects. Their results were sharpened by Kushner (1997). Carriere & Reinsel (1993) showed that if there are only two periods then the designs of Patterson (1952) are optimal for direct treatment effects; surprisingly, they did this without any reference to previous *Biometrika* papers on cross-over trials. Martin & Eccleston (1998) found variance-balanced designs for arbitrary patterns of within-subject correlation in the presence of residual effects.

9. Complicated block structures

Sets of experimental units with several different systems of blocks arise quite naturally in research in agricultural and related areas where there is heterogeneity: small blocks nested in large ones; rows crossed with columns; the three-way cross in a room for growing mushrooms (Preece et al., 1973); row–column designs with split plots (Yates, 1939); blocks which are each split into rows and columns; and so on.

When all the block effects are assumed to be fixed then treatment effects are estimated by projecting the data on to the bottom stratum, which is the space orthogonal to all types of block. Then all that matters is the information matrix L for the bottom stratum. An equireplicate design is said to be balanced if L is completely symmetric. *Biometrika* has published a sequence of papers on such balanced designs for blocks with nested rows and columns; see Singh & Dey (1979), Agrawal & Prasad (1982), Cheng (1986), Sreenath (1989) and Morgan & Uddin (1990).

On the other hand, the block effects may be random, so the block systems determine the pattern of the covariance matrix V. The concepts of orthogonality and balance among the block factors mean different things to different authors; various combinations of these give what is called 'orthogonal block structure', which may be characterised briefly by the fact that the eigenspaces of V depend on the pattern of its entries but not on their actual values. These eigenspaces are called 'strata'.

The general theory of designs with orthogonal block structure was worked out by Nelder (1965a, b). However, his work was overlooked by statisticians for several years, and other people reinvented the theory for various special cases.

In the most straightforward designs, the treatments subspace has a basis of contrasts each of which is estimable in only one stratum. Such designs are called orthogonal by some authors; they include the classic split-plot design and many factorial designs constructed from Abelian groups by the technique of confounding.

In the next simplest case, the information matrix in every stratum is completely symmetric. These designs are called balanced, but compare this with balanced designs for fixed block effects. Preece (1967) introduced nested balanced incomplete-block designs: small blocks are nested within large blocks, and treatments form a balanced incomplete-block design with respect to each type of block.

Orthogonality and balance, in the senses just defined, are both special cases of general balance, which is attained by a design if there is a common basis of eigenvectors for all the information matrices; that is, if all the information matrices commute with each other. Lewis & Dean (1991) explained this concept very clearly for row–column designs. Other special cases of general balance are partial balance in every stratum, for example the factorial balance in a row–column design discussed by Suen & Chakravarti (1985), and

so-called adjusted orthogonality, which was introduced by Eccleston & Russell (1975, 1977). Designs constructed by using Abelian groups in the first sense of generalised cyclic, such as those in John & Eccleston (1986), are generally balanced, while those which satisfy the second sense of generalised cyclic, such as those of Ipinyomi & John (1985), may not be.

The importance of orthogonal block structure and general balance is that the algorithms of Nelder (1968), Wilkinson (1970) and Worthington (1975) can be used to combine information across strata without ever needing to do numerical operations more complicated than sweeping or taking weighted averages. Moreover, essentially the same algorithms can be used for all orthogonal block structures; there is no need for one analysis for nested blocks, another for row–column designs and so on.

Within two decades of the publication of these algorithms there was no longer any need to avoid matrix inversion in routine computer programs. In a paper with much more wide-reaching importance than its title suggests, Patterson & Thompson (1971) gave a method for combining information which does not need orthogonal block structure or general balance. However, they did assume normal responses. Building on the work of Hartley & Rao (1967), they proposed maximising the likelihood of the projection of the data orthogonal to the treatments space. The method has come to be called REML or 'restricted maximum likelihood', although its authors call it 'residual maximum likelihood'; it is widely used.

The question of optimality for complex block structures has not been sufficiently investigated. If all block effects are fixed then we need good properties of the information matrix in the bottom stratum, just as we do for incomplete-block designs. John & Street (1992) investigated this for block designs with nested rows and columns and immediately ran into a problem that does not occur for block designs: should we restrict attention to generally balanced designs, whose algebra is relatively straightforward? A slightly surprising result discovered independently by several authors, starting with Bagchi et al. (1990), is that the best information matrix in the bottom stratum may come from a design that is very poor for one of the systems of larger blocks; an example in *Biometrika* was given by Altan & Raghavarao (1996).

However, when block effects are random then we may need good properties in more than one stratum. Morgan (1996) has shown that nested balanced incomplete-block designs are optimal among designs for nested blocks, but they do not exist for most sets of values of the parameters. To compare non-balanced designs we need to have at least a rough idea of the relative sizes of the components of variance. Examples are given by Morgan & Uddin (1993), Leeming (1997) and Bailey (1999).

The approach taken by Williams & John (1996) is to optimise for the largest blocks first, then optimise for smaller blocks within what has already been chosen. The rationale is that if smaller blocks are shown to be ineffective then they can be ignored in the analysis. Indeed, this is a second motivation for resolvable incomplete-block designs. However, Fisher and Yates's warnings about designing for a system of blocks with fixed effects and then ignoring it in the analysis are still valid. On the other hand, if the smaller blocks have random effects then ignoring them in the analysis amounts to replacing a negative, or small positive, estimate of a variance component by zero; Nelder (1954) explained why this is not recommended as a routine procedure.

10. Neighbours in time or space

10·1. *Changing treatments*

Cox (1954) pointed out that it may not be possible to make arbitrary changes of treatment from one period to the next. If the experimental units are just a line of subsequent

time periods then there is no solution to the problem. He gave a solution for what are effectively cross-over trials when the level of each treatment factor can only increase within each subject.

When the experimental units are just time periods then there may be scope for design if the treatments are factorial. It may happen that there is one factor A whose levels are difficult or expensive to change. Then adjacent periods must be grouped into blocks and levels of A applied to blocks, as in a split-plot design. In other factorial experiments every change of level of every factor is costly, so a design is sought which minimises the total number of changes of level: these designs are called Gray codes.

10·2. *Trends*

When experimental units form a long line in either time or space, they may affect the responses by a low-order polynomial trend. Although Hald (1948, § 15) had suggested that this might be true for agricultural fields, the problem of designing experiments for this situation was first considered in *Biometrika*, by Cox (1951). He observed that if the design is symmetric about its mid-point then all treatment contrasts are orthogonal to all odd-order orthogonal polynomials in the trend. Exhaustive search gives symmetric designs for small numbers of treatments in which treatment contrasts are orthogonal to the quadratic orthogonal polynomial and hence to the whole cubic trend. He continued by formulating what is now called A-optimality, the minimising of the average variance of treatment differences, and showed that designs orthogonal to the polynomial trend are A-optimal.

Snell & Bryan-Jones (1968) gave a design for a specific application in which treatment contrasts are nearly orthogonal to linear trend. More sophisticated methods have since been developed for ensuring that treatment contrasts are orthogonal to low-order polynomial trends; see Cheng (1990) and Jacroux & Ray (1990). For a factorial design, Bailey et al. (1992) showed that it is possible to adapt the confounding method in equation (1) to construct designs in which only high-order interactions are confounded with trend. Interestingly, Kobilinsky & El Mossadeq (1992) showed that the same method, but confounding low-order interactions, constructs Gray codes. Atkinson & Donev (1996) gave an algorithm for constructing trend-free designs which may not be either full replicates or regular fractions.

10·3. *Neighbour interference*

In a field trial, the treatment on any given plot may affect the responses on neighbouring plots, for example by shading them, stealing nutrients from them or passing disease on to them. The response on plot i might then be assumed to have expectation of the form

$$\lambda_{\tau(i+1)} + \delta_{\tau(i)} + \rho_{\tau(i-1)}, \tag{9}$$

where $\tau(i)$ is the treatment on plot i, δ_j is the effect of treatment j on the plot to which it is applied, and λ_j and ρ_j are the effects of treatment j on the plots immediately to the left and right of those to which it is applied.

When λ_j is assumed zero for all j then the neighbour effect is said to be one-sided. A special case occurs when experimental units are time periods and there are first-order residual effects. If every treatment is left neighbour to every treatment, including itself, equally often then the design is optimal. Usually this demands more replication than is available. Sometimes all distinct pairs occur as neighbours but treatments are not allowed

to neighbour themselves. Both types of design were called 'serially balanced' by Finney & Outhwaite (1955, 1956), and are now called 'neighbour-balanced at distance one'.

There are obvious modifications when the experiment is in several blocks, each of which is a long line or a circle. Rees (1967) gave neighbour-balanced designs for circular blocks, and Azaïs et al. (1993) adapted these for linear blocks with border plots at each end. These neighbour-balanced designs demand high replication, so designs for real experiments are frequently required to satisfy only that each pair of distinct treatments be neighbours at most once.

When both left- and right-neighbour effects are assumed nonzero then neighbour-balanced designs are often recommended but it is not obvious that they are optimal. Even if they are also neighbour-balanced at distance two, all that is guaranteed is that the information matrices for each of λ, δ and ρ are completely symmetric. Hardly anything is known about optimal design for three non-interacting factors, except in such special circumstances as those given by Stewart & Bradley (1991) and Druilhet (1999).

Equation (9) can be generalised to two dimensions, or to include plots further away. There are corresponding generalisations of neighbour-balanced designs, with many slightly different definitions. For example, Uddin (1990) combined neighbour balance with a complicated block structure.

Monod & Bailey (1993) considered two-factor designs in which only one factor has a neighbour effect. They gave constructions for designs in which each treatment is equally often neighbour to each level of the factor with a neighbour effect.

10·4. *Neighbour correlations*

Since Fisher's day, agricultural field trials have used blocks as a simple but approximate way of allowing for patterns in fertility. A few occasional papers suggested more realistic methods. Papadakis (1937), Atkinson (1969), Bartlett (1978) and Kempton & Howes (1981) proposed adjusting each response by the responses on neighbouring plots. In *Biometrika*, Williams (1952) proposed modelling the fertility changes by a low-order auto-regressive process. This caused him to recommend designs with undirectional neighbour balance at distance one, and possibly also at distance two: now each pair of treatments should be neighbours equally often in either order. He considered that fertility might also exhibit a fixed trend, so he proposed that his neighbour-balanced designs should consist of a sequence of complete blocks, possibly with a few border plots. Butcher (1956) made the idea of neighbour balance at distance d more precise: the number of times that a pair of treatments occurs on plots distance d apart should depend only on whether the treatments are equal. He also admitted that this property was convenient, in that it gives a completely symmetric information matrix for treatment effects, but may not be optimal.

Duby et al. (1977) made a numerical investigation of various designs under the assumption that correlation decays exponentially with distance. Kiefer & Wynn (1981) considered designs in linear blocks in which there is a correlation between nearest neighbours only. They concluded that designs with undirectional neighbour balance at distance one are optimal. They called these 'equineighbored'. Designs for other sorts of within-block correlation were considered by Gill & Shukla (1985), Kunert (1987) and Azzalini & Giovagnoli (1987).

However, it was only really with the discussion sparked by the paper read to the Royal Statistical Society by Wilkinson et al. (1983) that these ideas began to be taken seriously. *Biometrika* certainly contributed its share of papers proposing new models for responses

in field trials, and hence new methods of analysing data from such experiments; see Green (1985), Martin (1982, 1986) and Williams (1986). Many of these model the plot effect by correlation that decreases with distance. New methods of analysis impose new criteria for good designs. Designs with some sort of undirectional neighbour balance were recommended for these new methods of analysis by Atkinson (1969) and Uddin & Morgan (1997); Martin (1986) and Williams (1986) coupled neighbour balance with randomisation; Bellhouse (1984) concluded that very systematic designs should be used, in contradiction to Atkinson (1969) and Bailey, Azaïs & Monod (1995).

Just as we saw for orthogonal arrays in § 5, and trend-free designs in §§ 10·1–10·2, we find that a single combinatorial object, here a neighbour-balanced design, is a good solution to two different problems, those of neighbour interference and neighbour correlations. However, the problems remain quite distinct, and what is a good method of data analysis to deal with one is bad for the other.

A slight warning against the idea that the same design is good for different problems was sounded by Cheng & Steinberg (1991). They showed that where the trend affects a line of experimental units according to an autoregressive process then a factorial design whose main effects are orthogonal to low-order polynomial trend may not be very efficient; as Constantine (1989) had shown, it is also necessary to have many changes of level.

11. Treatment structures other than factorial

11·1. *Introduction*

Even for qualitative treatments, there are structures other than factorial. In each case the structure suggests suitable models to fit and important parameters to estimate. A design is good if it is tailored to these models or parameters.

11·2. *Half-diallels, intercropping and competition*

In diallel experiments in plant breeding, the treatments are crosses between males and females of, say, n parental lines. In the simplest case there is no self-cross and there is assumed to be no difference between males and females, so the treatments are effectively all unordered pairs from an n-set. This is called a half-diallel cross.

Yates (1947) suggested a simple additive submodel of the full model with $n(n-1)/2$ parameters: the response on pair $\{i, j\}$ has the form $\alpha_i + \alpha_j$, where α_i is called the general combining ability of line i. The remainder of the response, which is called the specific combining ability, is analogous to a two-factor interaction. If that 'interaction' is zero then the α_i can be estimated from a fraction of the pairs. Kempthorne & Curnow (1961), Curnow (1963), Fyfe & Gilbert (1963) and Hinkelmann & Kempthorne (1963) pointed out that such a fraction is combinatorially equivalent to an incomplete-block design with block size 2. However, in the diallel case the responses are effectively the block totals, so the best block design may not be the best diallel fraction, although if either design has all its canonical efficiency factors almost equal then so does the other. Mukerjee (1997) found some optimal fractions. Instead of a fraction we may have multiple replication in a block design. Then we effectively have a nested incomplete-block design; Gupta & Kageyama (1994) showed that the balanced designs of Preece (1967) are optimal. Triangular designs are a special sort of partially balanced design in which treatments are labelled by the unordered pairs from an n-set. It is natural to use such a design for a half-diallel with n parental lines, and Dey & Midha (1996) showed that they have good properties for estimating general combining effects. In fact, triangular designs have two canonical

efficiency factors, and the two sets of basic contrasts correspond exactly to general and specific combining effects. Thus Chai & Mukerjee (1999) have shown that triangular designs are good even when the specific combining effects are not assumed to be zero; this is the exact analogue of factorial balance for factorial design.

In intercropping experiments, we again have an unordered subset of an n-set. There may be a single response on each set, or a response on each item in each set. Combining effects of two, three, ... items are defined analogously to factorial effects. Federer & Raghavarao (1987) started to explore the problem of finding small fractional designs when high-order combining effects can be assumed to be zero.

Weighing designs are conceptually different from the above but abstractly very similar. For a spring balance, each weighing is a subset, possibly empty, of the objects. Thus again we have a block design where only the inter-block information can be used; Banerjee (1950, 1951) describes some examples.

In competition experiments we consider the case where blocks are pots or trees or animals: each treatment in the block has a direct effect on its own plot and a remote effect on all other plots in the block. As Pearce (1957) showed, half the parameters can be estimated within blocks, and the other half between blocks, so we need a block design that is efficient in both strata. This idea has been extended by Williams (1962) and McGilchrist (1965, 1967).

11·3. *Full diallels*

Yates also gave a family of models for the full diallel. It is appropriate whenever the treatments are all ordered pairs from two factors with the same set of levels, for example DNA and RNA from the same set of compounds (Alling, 1967). We do not know of any work on design for this structure.

11·4. *Control treatments*

Another type of treatment structure occurs when one or more of the treatments is a control, such as a placebo. In an important paper in *Biometrika*, Pearce (1960) defined an incomplete-block design to have supplemented balance if all the non-control treatments have the same replication, there is a single value for the concurrence between any pair of non-control treatments, and a, possibly different, single value for the concurrence between the control treatment and any other. There is an obvious generalisation to more than one control treatment.

Sometimes the control is included for demonstration purposes and the important comparisons are those among the other treatments. In other experiments, it is comparisons between the control and each other treatment that are of interest. A body of work has grown up on optimal designs for the second case, largely following Bechhofer & Tamhane (1981). Initially this was published in North America and ignored Pearce's work, but the optimal designs do frequently have supplemented balance and Pearce's contribution was eventually recognised; see Spurrier & Edwards (1986) and Gupta (1989).

Pigeon & Raghavarao (1987) extended these ideas to cross-over trials.

11·5. *Incomplete crossing*

Even factorials do not always have a completely crossed structure. If one factor is quantity of pesticide and the other is time of spraying then time is simply not defined for the zero quantity. This structure is best thought of as having a two-level factor to dis-

tinguish no pesticide from some pesticide: within the second level only, time and quantity are completely crossed. There are other situations where a complete cross may make logical sense but be impossible in practice. Gerami & Lewis (1992) considered designs for comparing each of two effective drugs with their combination when it would be unethical to include a placebo in the trial.

11·6. *Superimposed treatments*

Long-lived experimental material such as trees may have to be used for unrelated experiments in successive years. Then the first year's treatments may have a residual effect on the second year's responses, although it is usually assumed that there is no interaction with the second year's treatments. If the experiments are planned sequentially then the first year's treatments should be regarded as a system of blocks for the second year's experiment. If both experiments are planned together then we effectively have a factorial experiment for main effects only. Unlike factors in the designs in § 5, the two treatment factors may have large numbers of levels, possibly different.

Biometrika has played a leading role in giving designs for this specialised situation: Preece (1966) and Hedayat et al. (1970) required various sorts of balance and orthogonality; Hall & Williams (1973) used a cyclic construction.

12. Applications of optimal design theory

12·1. *Biometrika and 'applications'*

An application depends on the person and journal involved. To a pure mathematician an application might be 'the real numbers', whereas an industrial experimenter might expect an answer in terms of temperatures, pressures, catalyst concentrations and batches of reactants. The applications in this section are to designs for classes of problems less general than those of § 6. However, there are few *Biometrika* papers on specific applied problems.

12·2. *Response surface designs*

A typical response surface design would have as factors those mentioned above as characteristic of industrial experiments. The model will often be a second-order polynomial such as (7), in two or three factors; the response is expected to vary smoothly as a function of these factors.

'Response surface methodology' was developed and promoted by George Box as a result of his work at Imperial Chemical Industries. The methods include not only design but also the analysis of the experiments, the graphical representation of fitted surfaces and the experimental attainment of optimum conditions. Descriptions, in the chemical context, are given by Box in Chapter 11 of Davies (1956) and more recently by Box & Draper (1987).

Typically the experimental programme starts with designs for first-order models to screen the factors for importance; see § 5. Often these designs are 2^k factorials or their fractions. Box (1952) finds what are, in fact, D-optimum designs for up to $k-1$ factors in k trials, although the nomenclature of optimum designs was not established at that date. The optimum designs lie at the vertices of a regular $(k-1)$-dimensional simplex. He considers the rotation of this simplex design to avoid biases due to omitted second-order terms. Under plausible assumptions in the absence of specific knowledge about the omitted

terms, it is shown that the biases are unaffected by the orientation of the design. However, an example shows how to rotate the design to reduce biases due to the estimated effect of a trend in time.

Rotation of the simplex designs of Box defines a spherical design region. De Feo & Myers (1992) consider rotation of first-order designs within a cubical region. They are partially concerned with the detection of lack of fit and compare designs using the noncentrality parameter for omitted second-order terms. Unfortunately, they overlook Box's paper and reproduce some of his results.

The focus of response surface work is on the properties of second-order designs. The formulation of Box & Draper (1963) is canonical. There is an experimental region which includes a region of interest, in this case taken to be spherical, over which good prediction of the response is required. The model to be fitted is second-order, but it is desired to guard against the effects of omitted third-order terms. The mean squared error of prediction over the region of interest is then a function both of the average variance of prediction and of the squared bias. Box & Draper show that the bias is minimised if the moments of the design up to a certain order are the same as those of the region of interest. This leads to the selection of rotatable designs in which the variance of the response at x depends only on the distance of x from the centre of the design. The scaling of the design depends on the relative importance of variance and bias. Box & Draper commend all bias designs, which show only a relatively small increase in the average variance compared with those for which the variance is minimised. Similar results for first-order models were presented earlier by Box & Draper (1959).

The designs which result from these calculations are central composite designs; that is, they consist of the points of a 2^k factorial, augmented by a few centre points, the number being determined by the desire for approximate rotatability, and by star points at which all factors except one take the values at the centre of the design. Similar designs are investigated by Draper & Lawrence (1965) for cuboidal regions of interest, for which the star points are situated further from the centre of the design than they are for the spherical region. Draper & Lawrence (1966) look at the slight consequences of using designs for spherical regions when the region is cuboidal and vice versa. Rotatable designs for estimating tensors are described by Hext (1963).

The Box & Draper formulation is for a set of smooth alternatives to the model being fitted, typically third-order polynomials. Welch (1983) uses optimum design theory to consider a more general class in which perturbations of $\pm\delta$ are added to the response at each corner of a cubic design region. As the variance of δ increases, the criterion produces a family of designs moving from the first-order 2^k factorial to a uniform design over the design region. This corresponds to the complete swamping of the model by the perturbations. Steinberg (1985) uses a Bayesian approach to the scaling of the factors in 2^k designs when, again, the model is not adequate.

If the model is not adequate it is not obvious that least squares estimators are optimum. Kupper & Meydrich (1973) look at biased estimators which are an affine transformation of the least squares estimators. Atkinson (1970) gives designs for the slope of a response surface when there may be biases from omitted quadratic terms. The designs found by Mukerjee & Huda (1985) are optimum for the slopes of correctly specified models including interactions.

In practice it is not always certain whether or not all experimental variables will affect the response. In the selection of regression variables using Mallows's C_p (Mallows, 1973) there is a trade-off between the increase in variance from including an extra variable and

the reduction in bias. Davies (1969) applies related reasoning to the inclusion or exclusion of a single variable in a 2^k factorial.

The purpose of a response surface design is to provide a set of experimental conditions in a k-dimensional space which makes it possible to fit a low-order polynomial model. In the Box & Draper formulation it is assumed that the region of interest is known, when a design is obtained which is scaled by the region, some design points lying within the region and some outside. When optimum designs are used the experimental region is specified. Design points cannot lie outside this region, although many lie on the boundary. The results on second-order designs of Farrell et al. (1967) show that the sets of design points for optimum designs lie on shells of points with $0, 1, 2, \ldots, k$ nonzero coordinates, the extension of the sets of points for model (7) to k factors. Apart from the interpretation of the region, the D-optimum designs are very similar to those of Box & Draper. Atkinson (1972) finds exact D_s-optimum designs for second-order terms when it is required to check whether a first-order model holds. Tables of small exact D- and D_s-optimum designs are given by Atkinson (1973) and by Pesotchinsky (1975). Lucas (1977) calculates D-, A-, G- and E-efficiencies of a four-factor design as the weight on the centre point changes.

A final, quirky contribution is that of Box & Draper (1975) who list 14 criteria that a response surface design should satisfy. They focus on robustness, which is interpreted as the absence of leverage in any observations. Thus all points should have the same variance of prediction, a condition, in the continuous theory, satisfied by D- and G-optimum designs. This conclusion contrasts oddly with Box & Draper (1987, Ch. 14), which repeats the 14 points and describes ways in which optimum design theory fails to capture the essence of designing experiments to estimate response surfaces in which the model is only approximate. The main technical problem which results is that of scaling optimum designs.

These contributions are the tip of the iceberg, the body of which is the rejection of Kiefer's work by many applied and industrial statisticians in the United States. Egon Pearson (1968, p. 456) refers to the stress that such battles as those between his father and Fisher caused the participants. There must have been some lively mealtimes at Well Road. However, he does not mention the damage that such disagreements can do to the development of science, in this example of experimental design not just to statistics but to those technological subjects relying on the results of planned experiments. A typical example of such intellectual confusion is Myers & Montgomery (1995, § 8·4) who set up a false dichotomy between 'response surface designs' and 'alphabetic optimality' applied to the same models. Despite the close relationship between the answers provided by the two formulations, Kiefer's work is criticised because of the scaling of the design region described above and because it fails to give a satisfactory distribution of variances of prediction. Unthinking application of computer implementations of optimum design is deprecated, as though unthinking implementation of any design procedure will not lead to poor results. A mathematical comparison between the two approaches is in Kiefer (1961), particularly § 3·2, with further numerical comparisons in other papers collected in Brown et al. (1985).

12·3. *Algorithms*

A strength of optimum experimental design is that there is a clearly defined criterion which can be numerically maximised to provide a design. One example is in response surface designs for irregular regions, some examples of which are given in Chapter 16 of

Atkinson & Donev (1992). It would be hard to infer these designs from the structure of rotatable designs on regular regions.

Little of the work on algorithms for designs has appeared in the major statistical journals. Vuchkov (1977) provides an algorithm for calculation of singular designs by a regularisation of the design matrix in which a small multiple of the identity matrix is added to the information matrix. This can be useful at the start of the iterative construction of a design which is nonsingular when enough trials have been added: the regularisation is successively reduced as trials are added. It is also useful in the numerical construction of, for example, singular c-optimum designs.

One problem in the response surface methods is how to block the designs. Atkinson & Donev (1989) provide algorithms for blocking response surface designs. Their exchange algorithm is also useful for the construction of exact D-optimum response surface designs and for finding designs in which there is a combination of discrete and continuous factors. Similar methods were proposed at the same time in *Technometrics* by Cook & Nachtsheim (1989). Methods for the construction of continuous minimax designs are presented by Wong (1992).

In algorithms such as that of Atkinson & Donev (1989) an explicit formula is available for the effect on the determinant of the information matrix of removing a design point and replacing it with another. A problem with similar algorithms for the incomplete-block designs of § 7 is that several treatments have to be interchanged at once. Williams & John (2000) use what is sometimes called the 'Sherman–Morrison–Woodbury' formula (Harville, 1997, p. 423) to derive an explicit expression for the effect of interchange in α-designs which allows use of the average efficiency factor as a measure of design optimality. The use of surrogate measures of performance is avoided.

12·4. *Mixtures*

From the algorithmic point of view, the design of mixture experiments is not very different from that for response surface designs. The models are usually polynomial, rewritten to satisfy the constraint $\sum x_i = 1$. An elegant, symmetrical form of the models is given by Scheffé (1958, 1963), together with simplex–lattice and simplex–centroid designs when experiments can be performed over the whole simplex defined by allowing each variable to vary from 0 to 1, provided the mixture constraint is satisfied. Optimum design algorithms provide designs when there are further constraints on the region, so that the experimental region is only part of the regular simplex. Two examples are given in Atkinson & Donev (1992, § 16·3) and a book-length treatment of mixture experiments by Cornell (1990).

A worrying feature of both Scheffé's designs and the D-optimum designs for irregular regions is that they contain many trials on the boundary of the design region. Unfortunately, the response surfaces for mixtures are sometimes highly curved near the pure components. An obvious example is that the properties of pure sulphur, saltpetre and charcoal are very different from their properties when blended in the proportions to make gunpowder. Response surface designs allowing for model inadequacy are found by Becker (1970) when the fitted model is first-order.

Cox (1971) is concerned with difficulties in the interpretation of the polynomial models of Scheffé. As well as the over-reliance on the properties of pure components and binary mixtures, there is the difficulty that two blocks of an experiment for which the response varies only by an additive constant will have different values of all the parameters. He

suggests models based on a standard mixture, for which the parameters have more nearly conventional interpretations. For full models the D-optimum designs for these models would be the same as those for Scheffé's polynomials, since these designs are invariant to affine transformations. However, the models are such that the vanishing of sets of components has a physical meaning. As far as we are aware, Cox's last sentence, 'the efficiency of various designs for the estimation of parameters remains to be investigated', still stands.

12·5. *Nonlinear models*

An elegant, early contribution to design for nonlinear models is Fisher (1960, § 68) who justified the use of the dilution series for estimation of a Poisson parameter when the response is the presence or absence of organisms; see Cox & Reid (2000, § 7·6). Chernoff (1953) introduced locally optimum designs in which the design may depend upon the unknown values of the parameters. Neither of these papers appeared in *Biometrika*, unlike the seminal paper of Box & Lucas (1959) which considers design for parameter estimation in some nonlinear models arising in chemical kinetics. If there are two consecutive first-order reactions $A \to B \to C$ and the concentration of B is measured at time t then the statistical model is

$$Y = \eta(t, \theta) + \varepsilon = \frac{\theta_1}{\theta_1 - \theta_2} \{\exp(-\theta_2 t) - \exp(-\theta_1 t)\} + \varepsilon, \tag{10}$$

where θ_1 and θ_2 are the rates of the two first-order reactions and the errors ε satisfy second-order assumptions. This, then, is an example of the nonlinear models for which the derivatives $\partial\eta(t, \theta)/\partial\theta_j$ in (5) depend on the values of θ.

Box & Lucas (1959) use Taylor series expansion of the model about the prior point values θ^0 to obtain locally D-optimum designs for parameters. These designs put measurements of the concentration of B at two times. It is a frequent phenomenon in designs for such nonlinear models that the number of design points is equal to the number of parameters. There is thus no way of checking the model.

There are several extensions of the results of Box & Lucas's paper. Since there are three chemical species, it might be possible to measure more than one at any time. Draper & Hunter (1966) use noninformative priors to obtain a design criterion for multiresponse experiments. This is the determinant of a sum of information matrices, weighted by the variances and covariances of the responses. Draper & Hunter (1967a) revert to experiments in which one response is measured but now include prior information, which adds to the information matrix (5) a quantity which does not depend on the design, but can be thought of as representing previous experiments. The last of this series of papers, Draper & Hunter (1967b), extends the use of informative priors to multiresponse experiments.

These papers are concerned with estimation of all the parameters in the model. If only a subset of the parameters is of interest, the appropriate criterion is D_s-optimality. Mike Box (1971), the nephew of George, derives D_s-optimality for nonlinear models. As in Draper & Hunter (1966), the criterion applies to measurements of multiple responses. He sketches an example in which three possible mechanisms are given for a chemical reaction. Only one of the mechanisms can be true. The models are not nested, so direct choice between them would require the methods for discrimination between models of § 6·4. However, the models are all special cases of a more general model, so that his D_s-optimum designs can be used to determine which submodel holds. Atkinson (1972) considers the

use of D_s-optimum designs for checking nonlinear models and discusses three ways in which a nonlinear model can be embedded in a more general family.

All of this work is concerned with deriving design criteria and designs. There is no attempt to derive the relevant equivalence theorems or make use of the results for checking the optimality of designs. An equivalence theorem for D- and D_s-optimality is given by White (1973) for point priors. She emphasises that the results apply not only to nonlinear regression models but also to generalised linear models. The full theorem for nonlinear models including proper prior distributions of the parameters is given by Chaloner & Larntz (1989), who exemplify designs for binomial models.

Several papers consider nonnormal models. Andrews & Chernoff (1955) find a locally optimum design when the dose is uncertain. In their example the dose is the Poisson-distributed number of bacteria given to an animal, the probability of a response depending on the number of bacteria administered. The two-parameter models for which Dette & Haines (1994) find E-optimum designs include binary responses. In Hoel & Jennrich (1979) the nonnormal response is the probability of an animal developing cancer in low-dose carcinogen testing. Since maximum likelihood estimation reduces, as it does for generalised linear models, to iteratively reweighted least squares, the asymptotic design problem transforms into one for heteroscedastic regression.

There are other ways in which these methods may require extension, either because of too severe approximations or to wider classes of models, as follows.

1. The methods for nonlinear models rely on a linearisation of the model and the assumption that the quantities of interest are functions of the asymptotic normal-theory confidence ellipse for the parameters, or their Bayesian equivalent. However, these contours may be far from elliptical. O'Brien (1992) uses a quadratic approximation due to Hamilton & Watts (1985) to find designs which have $p+1$ design points, where p is the number of parameters in the model. Such designs allow for model checking, although not necessarily in an optimum way.
2. It is assumed that the parameters are fixed; even in the Bayesian theory the parameters are usually taken to be sampled from the prior once for all experimental units. Mentré et al. (1997) design for random effects nonlinear regression models arising in toxicokinetics; D-optimality is used to find designs for the fixed effects and for the variances of the random effects.
3. Time has been ignored in the design of many of these experiments. For example, the Box & Lucas (1959) experiment for the intermediate product model (10) specifies the time at which an observation is to be taken. The next observation, taken at the same or a different time, is from another run on the plant and so provides an independent observation. However, in many processes the concentrations could be continuously monitored, when time series data would result. The experimental variables might be the temperatures and pressures often associated with response surface designs, which could vary during the course of the experiment. Partial solutions of this design problem are to be found in the chemical engineering literature; see, for example, Lohmann et al. (1992).

12·6. *Nonlinear aspects*

Designs for nonlinear models depend on the unknown values of the parameters, but such nonlinear problems can come from linear models. A very simple example is the

quadratic model

$$y = \beta_1 x + \beta_2 x^2 + \varepsilon, \tag{11}$$

when interest is in estimating the parametric function $g(\theta) = -\beta_1/(2\beta_2)$, the value of x at the maximum or minimum of the quadratic. A more complicated example is the response surface in several factors considered by Box & Hunter (1954), where interest is in estimating the maximum, a major feature of response surface methods. They unexceptionally use the Fieller–Creasey theorem (Fieller, 1940) to obtain an approximate confidence region for the maximum. However, they use the properties of the fitted surface to design a second block of experiments and then use the same method to find a confidence region for the combined results. This procedure fails to allow for the influence of the responses to the first design on the second design and consequently on inferences from the combined design.

Inference from a sequential design is considered by Ford & Silvey (1980), who add one trial at a time to a design to minimise the variance of the estimated maximum in (11). They find that the sequential design converges to the optimum design for known parameter values. They also find, from a simulation study, that the adaptive construction of the design has little effect on the properties of the confidence region for the maximum.

Ford et al. (1985) consider several examples in which the adaptive design depends on the observations and show that interval estimation based on the achieved design may be incorrect. Their suggested procedures do not readily generalise. Wu (1985) proves that, asymptotically, the adaptive nature of the design of Ford & Silvey (1980) does not affect the confidence interval for the maximum. He also considers the generalisation to a smooth q-dimensional function of the parameters of a normal-theory linear model. The convergence of the parameter estimates and the validity of the fixed design confidence region are established under conditions on the asymptotic behaviour of the eigenvalues of the information matrix. Hu (1998) summarises this and much related work and proves the strong consistency of Bayesian estimators of the parameters. In all this work it is sometimes difficult to verify that the conditions required, for example on the evolution of ratios of eigenvalues, are satisfied by a specific design procedure.

12·7. *Generalised linear models*

The expression of maximum likelihood estimation for generalised linear models as iteratively reweighted least squares reduces the design problem for these models to that of design for linear models with weights which are a function of unknown parameters. The equivalence theorem of Chaloner & Larntz (1989) was mentioned above. Introductory examples are given by Atkinson & Donev (1992, § 22·5). Ford et al. (1992) provide canonical designs for experiments with a single variable x. Since the weighting depends on x, the design problem can instead be treated as unweighted design over an induced design region in variables z. Examples of designs for the logistic model for binary responses with two factors are given by Atkinson (1995) who shows the dependence of the number of support points of the design on the values of the parameters of the linear predictor. Sitter & Torsney (1995) investigate multifactor designs and obtain two-level factorial designs in the induced design space for a linear predictor which contains first-order terms. The design weights are not the same for all design points.

None of this work is in *Biometrika*, but the papers that have appeared in this journal are concerned with this factorial structure. Cox (1988) argues that, for generalised linear models with canonical parameterisation and small effects, standard normal-theory results apply for local designs, yielding, for example, two-level factorials. He also considers the

effect of stratification, particularly for the comparison of two treatments for binary data. Randomisation, as opposed to balancing, induces an average loss of one observation per stratum. This is the analogue of the asymptotic results for normal-theory models in Cox (1957) which arise in calculation of the losses in sequential clinical trials mentioned in § 13.

The factorial designs found by Burridge & Sebastiani (1994) depend, for their optimality, on the values of the parameters in the linear predictor. They look at models with a power link and with variance proportional to the square of the mean. Under some conditions they find that designs consisting of points with just one variable at the high level and all others at the low level, plus one trial at the low level of all factors, are optimum. For large values of β_0 the support points of the design extend to include points of the 2^k factorial, which are not equally weighted. Brown (1973) finds designs for binary key models in which the binary response is biological activity, often observed without error, which depends on the presence or absence of particular molecular structures. There is a relationship with the error-correcting codes investigated in connection with two-level factorials in § 5.

Finally, there are two papers, Chernoff & Haitovsky (1990) and Zelen & Haitovsky (1991), in which there are two binary populations which are measured with error depending on the outcome. Design possibilities include checking all responses of one kind to see if they were subject to error, where checking may have a different probability of error from the original measurement.

13. SEQUENTIAL CLINICAL TRIALS

Patients arrive sequentially and are to be allocated one of t treatments. In 'sequential' designs the treatment allocations are made in the knowledge of each patient's prognostic factors but in the absence of any results on the effect of treatment. In 'adaptive' or 'data-dependent' designs allocation is made in the light of at least some earlier responses. The papers that have appeared in *Biometrika* cover aspects of both sequential and adaptive designs. In the latter case it is always assumed that information is available from all previous allocations.

The celebrated paper of Efron (1971) introduced biased-coin designs into the sequential design of clinical trials. There are two treatments and no prognostic factor. Patients arrive and are given one of the treatments and it is not known when the allocation of treatments will stop. If the design is unbalanced, that is, if the numbers of patients receiving the two treatments are not at all equal, the variance of the estimated treatment difference will be unnecessarily high. The allocation of treatments needs therefore to be reasonably balanced throughout the course of the trial, so that the trial cannot stop at a moment of appreciable imbalance. However, there also needs to be sufficient randomness that the allocation is not biased by the doctor's ability to guess the next allocation. Efron's design allocates the under-represented treatment with a probability between $\frac{1}{2}$ and 1, for example $\frac{2}{3}$. This paper sparked much work. Wei (1978) investigates random allocation and selection bias for designs in which the number of patients is fixed. Steele (1980) proves a conjecture of Efron that biased-coin designs are asymptotically free from accidental bias caused, for example, by secular trends in response. Atkinson (1982) extended the biased-coin design to any number of treatments and covariates, using optimum design theory to generate the allocation probabilities. An authoritative comparison of the asymptotic properties of the various procedures is Smith (1984), with simulations for small-scale properties in Atkinson (1999).

The sequential probabilities can alternatively be conceptually generated from an urn,

to which balls are added according to the previous allocation. Several sequential urn designs due to Wei are compared in Wei et al. (1986). Wei (1988) describes the extension of urn designs to be adaptive, by the addition of balls to increase the probability of continuing with a successful treatment. Begg (1990) contains a discussion by several statisticians of the analysis of an experiment generated according to such an urn scheme.

In many of the other papers on clinical trials the interest is at least as much in stopping and analysis as in the pattern of treatment allocations over time. Bias is usually ignored. For example, Pocock (1977) is concerned with using significance tests to decide when to stop a group sequential trial in which normally distributed observations are continually becoming available. The results only affect the design by whether or not the trial continues to the next group.

Often the response is not normal, but binomial, perhaps the success of a treatment or survival for at least a given time. Bandit processes provide a useful model for such clinical trials, the simplest being for the comparison of two treatments with binary responses. Sobel & Weiss (1970) consider play-the-winner rules for two binomial populations, when interest is not only in selecting the better population but also in using the inferior treatment as little as possible. Glazebrook (1978) uses the bandit indices invented by Gittins to extend the two-arm bandit approach to more than two treatments and more than two outcomes. There is unfortunately no application of the papers of Chernoff & Haitovsky (1990) and Zelen & Haitovsky (1991) on non-sequential designs for binary treatment subject to error, see § 12·7, to these adaptive design problems.

These bandit models assume that the treatments are known. Schaid et al. (1990) develop two-stage screening designs for survival comparisons of new treatments with a control. Only treatments which are promising in the first stage are taken through into the second. Measurements are made at only two time periods and the decision is which new treatments, if any, should continue to the second stage. Boys et al. (1996) consider an earlier stage of drug development in which there are a number of compounds to be screened for therapeutic activity and toxicity. There may be several screens of both types, and the design problem is to order the screens. Robinson (1978) also considers toxicity, but is concerned with a single compound, for which sequential designs are used to find the maximum dose which can be administered, subject to a restriction on toxicity. A review of more recent work in this area is Rosenberger (1996).

Disease detection is different from clinical trials in that no treatments are to be compared, the design problem being solely when screening should take place. Hu & Zelen (1997) investigate the properties of models for early detection programmes for cancer. Time is also the only factor in the designs of Box & Lucas to estimate the parameters in the intermediate product model (10). The solution to the design problem of when to take measurements was there determined by prior information about the values of the parameters. In the absence of such prior knowledge, Bergman & Turnbull (1983) provide an adaptive scheme of sampling for destructive testing when the failure time is known to be exponential and the purpose of the experiment is to estimate the value of the single parameter of the distribution.

14. Future prospects

Tippett, in the discussion of Yates (1935) states as follows.

> It was a little hard at first for one brought up in a school of physics to abandon the

> sacred principle of varying one factor at a time but . . . it certainly led to an efficient use of experimental resources if all . . . [factors] . . . were investigated at the same time.

Two-thirds of a century later this particular battle remains won in agriculture, but less surely so in industry.

When designs created for one area of application are used in another, typically there are two developments. The first is that the new area adopts design principles that have been accepted in the other for some time. Examples include not only factorial design but also the split-plot principle, which has yet to be recognised seriously in environmental toxicology. The second is that the slightly different requirements of the new area make different demands on the design, and so the theory and practice of design is itself modified. A recent example of this is experiments on training people to do certain computer tasks. The experiments are cross-over trials, but the model includes experience, which is confounded with time period, as a fixed effect which may interact with task. Thus good designs will not necessarily be the same as those described in § 8. The future of design is heavily linked to its applications.

One currently active area is the improved analysis of experiments with complicated variance structures. Kenward & Roger (1997) show how to improve the estimators of the variances of treatment contrasts when REML is used for structures such as those described in §§ 7 and 9. There is continuing debate over the best analysis of experiments laid out in space; see § 10 and the Bayesian perspective in Besag & Higdon (1999). As new methods of analysis become adopted, knowledge will become needed about which types of design are good for those analyses. Probably such knowledge will come in two stages: first a combinatorial condition, such as neighbour balance, or numerical criterion, such as A-optimality, that tends to make a design good; then some direct methods for constructing such designs and algorithms for the cases when the direct methods are not applicable.

One important industrial topic not covered in *Biometrika* is the design of experiments for computer simulation, where experiments are performed directly on the code, for example by changing parameters in a complex, but error-free, simulation which could involve the numerical solution over time of sets of differential equations. References to adaptive and space-filling designs for this purpose are in Bates et al. (1996).

The use of computers in designing experiments will certainly increase, most obviously in the numerical construction of optimum designs for increasingly complex situations. For example, Martin et al. (2001) find designs for mixtures where the experimental region is subject to a variety of linear constraints. More computationally challenging are designs for time profiles of process variables mentioned at the end of §12·5.

Intellectual challenges will continue to be provided by adaptive designs, several kinds of which are described in the papers collected in Atkinson et al. (2001). Pronzato (2001) describes a problem in which units arrive with a known value of an explanatory variable and the sequential decision is whether or not to take a measurement. The numbers of units and of allowed experiments are both known. This problem is related to the design of sequential clinical trials discussed above. Kpamegan & Flournoy (2001) and Hardwick & Stout (2001) extend the problem to that of finding a dose which is simultaneously efficacious and has low side-effects, the experimental programme to involve as few patients as possible receiving a suboptimal dose. Some of the mathematics involved is related to that for bandit processes.

Other topics in an early stage of development which have not been discussed here include designs for training neural networks (Titterington, 2001) and combinatorial chem-

istry (Bond et al., 2001). In the latter, the ability to perform large numbers of experiments on microscopic amounts of material has led to very large experiments: the use of the standard ideas of fractional replication might lead to significant savings in time and cost.

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